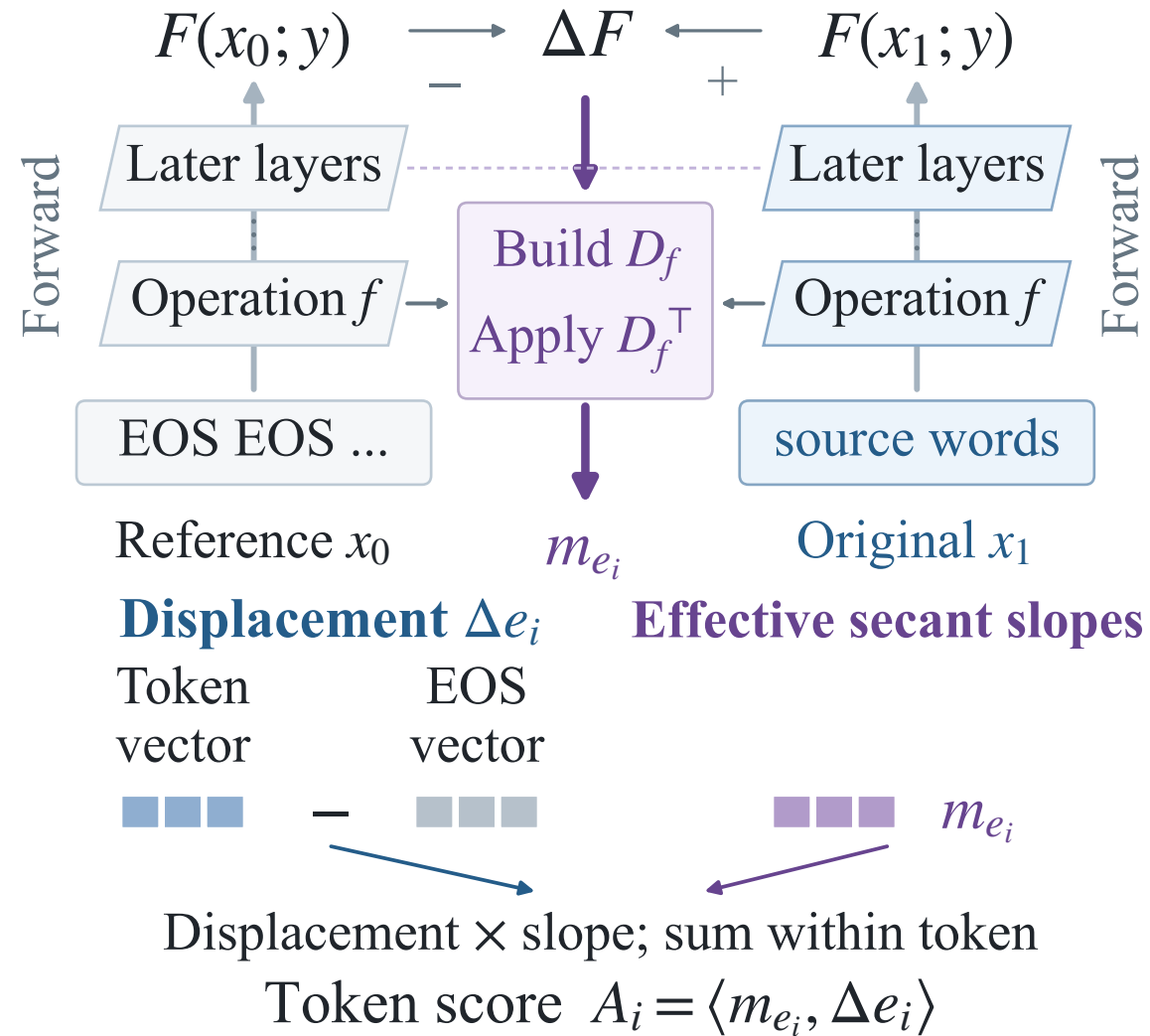


(a) Overall method

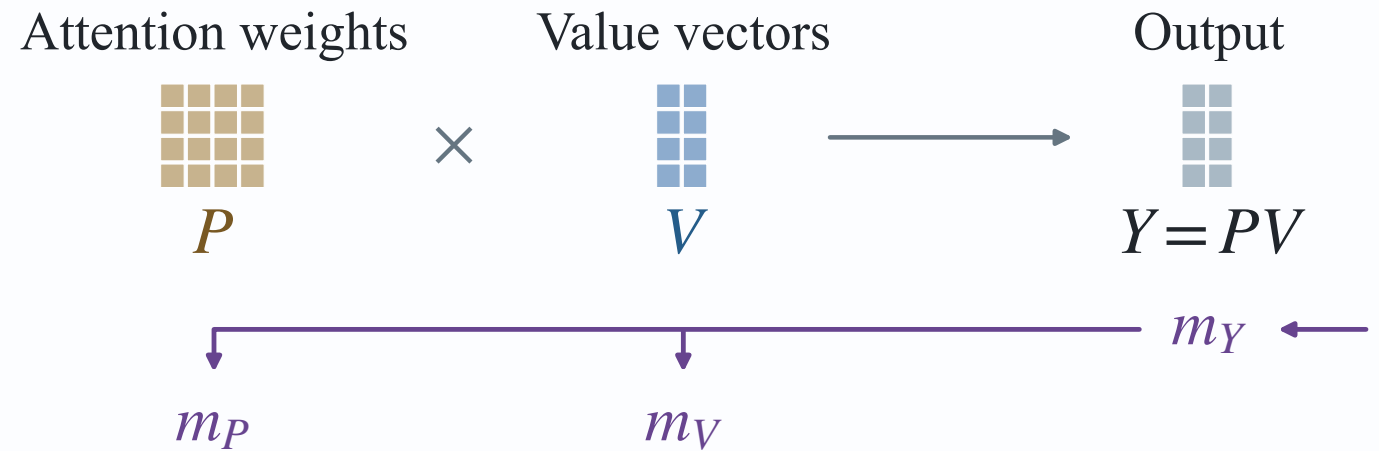
Two forward runs with fixed response y



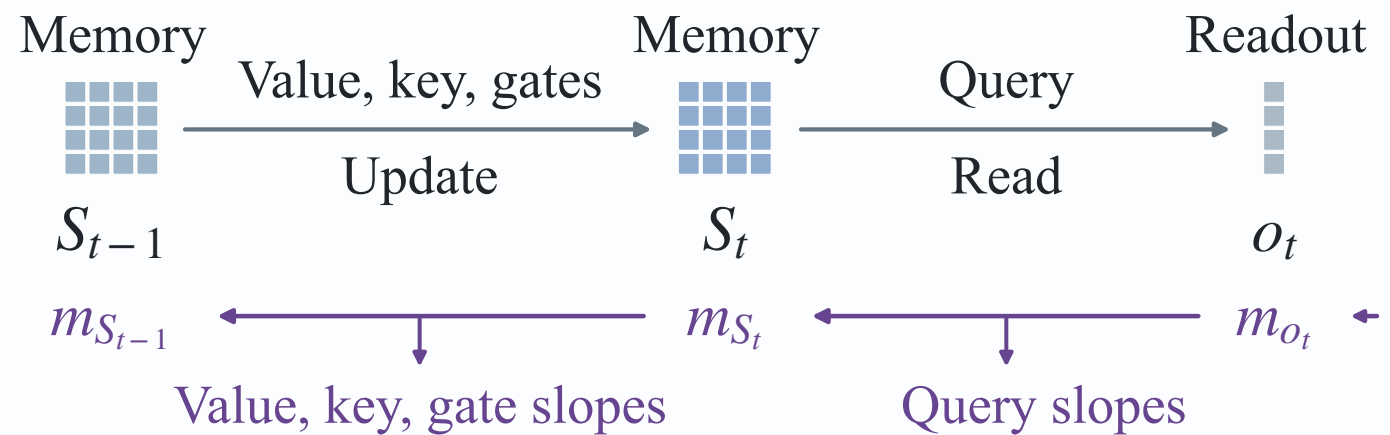
Reverse-pass examples

→ Forward ← Backward $D_f^\top$

(b) Attention product



(c) Gated DeltaNet



(d) Choose the playwright, not the composer.

CONTEXT EXCERPTS

... William Shakespeare's play

... The score is by William Walton.

REQUESTED ROLE (EXCERPT)

... the author of the play...



Name-span total (nats)

+1.56

-1.13

Token contribution (nats)

≤ -2.5

0

$\geq +2.5$

Shown spans underlined

MoreHopQA example 1

Answer

William
Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources

(a) Qwen3-8B

Context

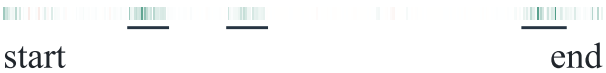
One of the special magic numbers for deafening-opium is: 5443951.

...
One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

Full input



(b) Qwen3.5-9B

Context

One of the special magic numbers for deafening-opium is: 5443951.

...
One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

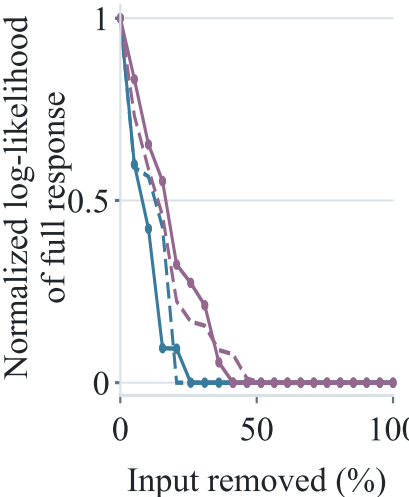
Full input



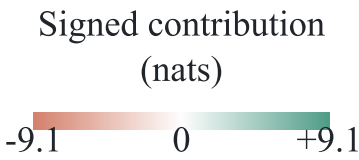
(c) Deletion

Answer

5443951, 8698256



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT



(a) Qwen3-8B

Context

Townsend Putnam Coleman
III (born May 28, 1954) is
an American voice actor
...
he also did additional
voices in films "Fantasia
2000" (1999) and "Sing"
(2016)
...
It was directed and
written by Garth Jennings,
co-directed by Christophe
Lourdelet,

Question

How many consonants are
there in the last name of
the person who directed
and wrote the 2016 film
featuring the voice of
Townsend Coleman?

Full input



(b) Qwen3.5-9B

Context

Townsend Putnam Coleman
III (born May 28, 1954) is
an American voice actor
...
he also did additional
voices in films "Fantasia
2000" (1999) and "Sing"
(2016)
...
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Question

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Townsend Coleman?

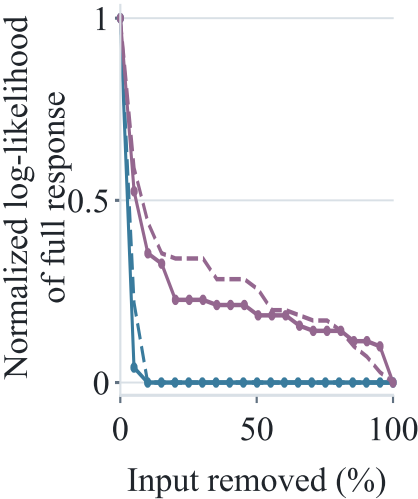
Full input



(c) Deletion

Answer

6 consonants
in Jennings



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT

Signed contribution
(nats)

